

Chapter 4

Standard Cost and Variance

Static Budget, Flexible Budgets and Variance Analysis

Learning Objectives

- I. Standard Cost(Budgeted rate) , standard setting and Variance*
- II. Static budget and performance report using static budget(Level-1 variance analysis)*
- III. Prepare flexible budgets and performance reports using flexible budgets(Level-2 variance analysis)*
- IV. Compute variances for direct materials and direct labor(Level-3 variance analysis)*
- V. Compute variances for manufacturing overhead(Level-3 variance analysis)*

Learning Objectives-I

*Standard Cost(Budgeted rate) , standard setting and
Variance*

Standard Costs

Standards are:

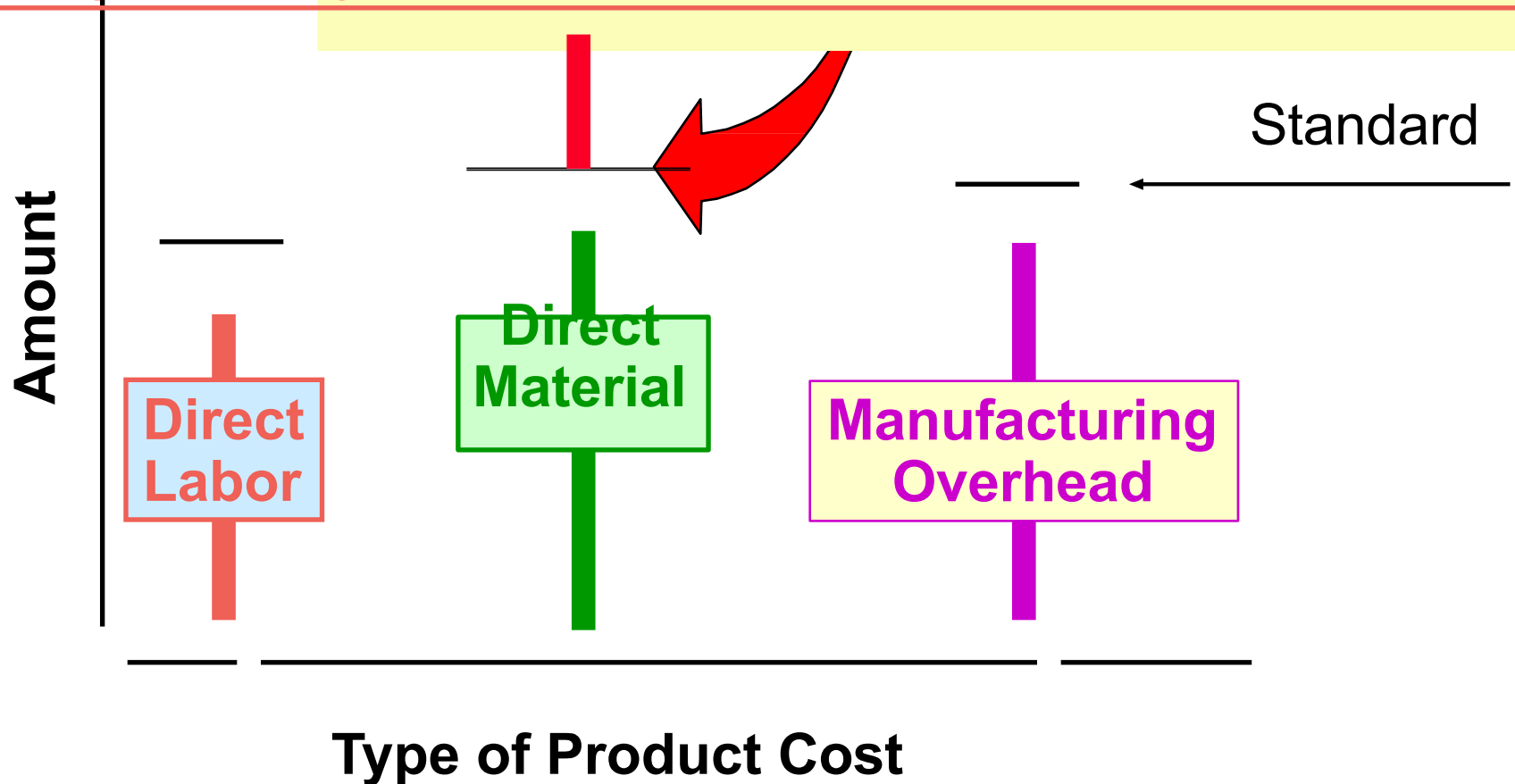
- Benchmarks or “norms” for measuring performance.
- Standard Cost has two component :

Quantity standards specify how much of an input should be used to make a product or provide a service.

Rate (price) standards specify how much should be paid for each unit of the input.

Standard Costs Cont..

Deviations from standard deemed significant are brought to the attention of management, a practice known as **management by exception**.



Setting Standard Costs

Cost Standards are set by:

- Accountants,
- Engineers,
- Purchasing agents, and
- Production managers combined efforts to encourage efficient future production.



Engineer

Managerial Accountant

Setting Direct Material Standards

Price Standards

**Final, delivered
cost of materials,
net of discounts.**



Quantity Standards

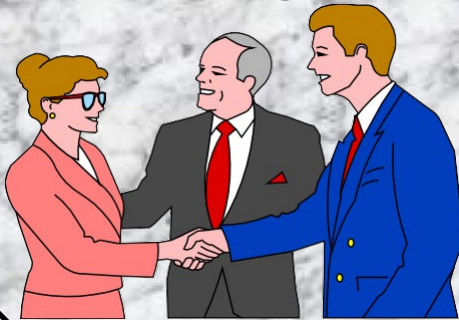
**Summarized in
a Bill of Materials.**



Setting Direct Labor Standards

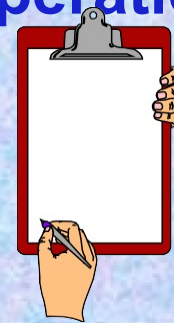
Rate Standards

Often a single rate is used that reflects the mix of wages earned.



Time Standards

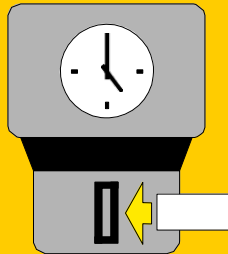
Use time and motion studies for each labor operation.



Setting Variable Overhead Standards

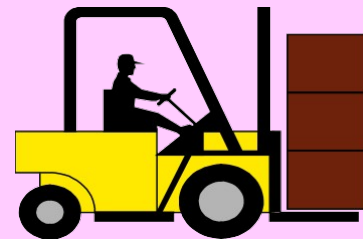
Rate Standards

The rate is the variable portion of the predetermined overhead rate.



Activity Standards

The activity is the base used to calculate the predetermined overhead.



Example: Standard Cost of Variable Product Cost

A standard cost for one unit of product(FG) may look like this:

Inputs	A	B	A x B
	Standard Quantity or Hours	Standard Price or Rate	Standard Cost per Unit
Direct materials	3 unit	\$ 4.00	\$ 12.00
Direct labor	2.5 hours	per unit 14.00	35.00
Variable mfg. overhead	2.5 hours	per hour 3.00	7.50
Standard unit cost			\$ 54.50

Standards vs. Budgets

- Are standards the same as budgets?

- ❖ A budget is set for total costs and a standard is a per unit cost.

- ❖ Standards are often used when preparing budgets

- ❖ Therefore, when variance analysis is made actual result is compared against budget prepared using standard cost.

Price and Quantity Standards

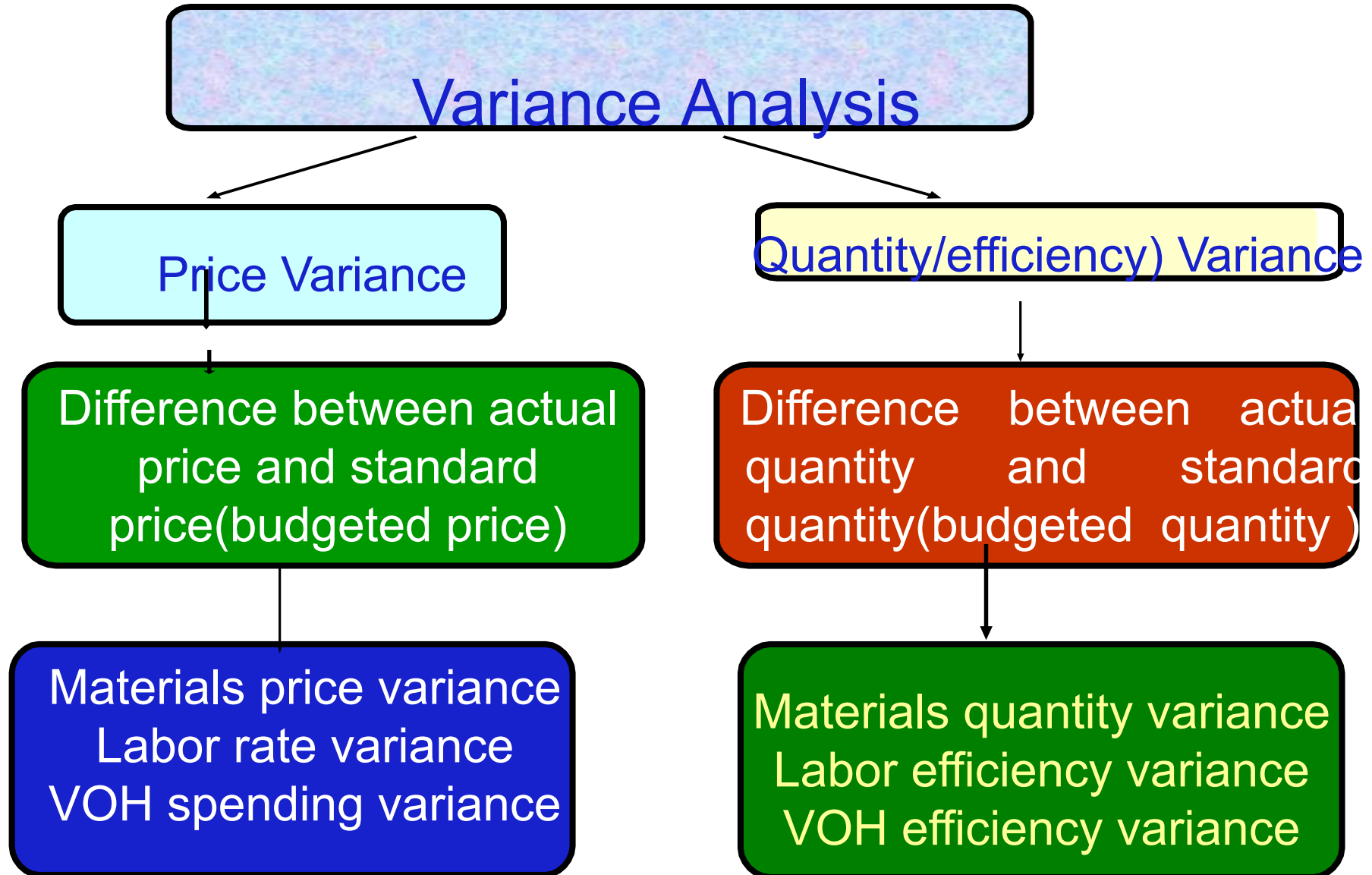
Price and quantity standards are determined separately for two reasons:

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graph TD; A[Price and quantity standards are determined separately for two reasons:] --> B[1 The purchasing manager is responsible for raw material purchase prices and the production manager is responsible for the quantity of raw material used]; B --> C[2 The buying and using activities occur at different times. Raw material purchases may be held in inventory for a period of time before being used in production.];
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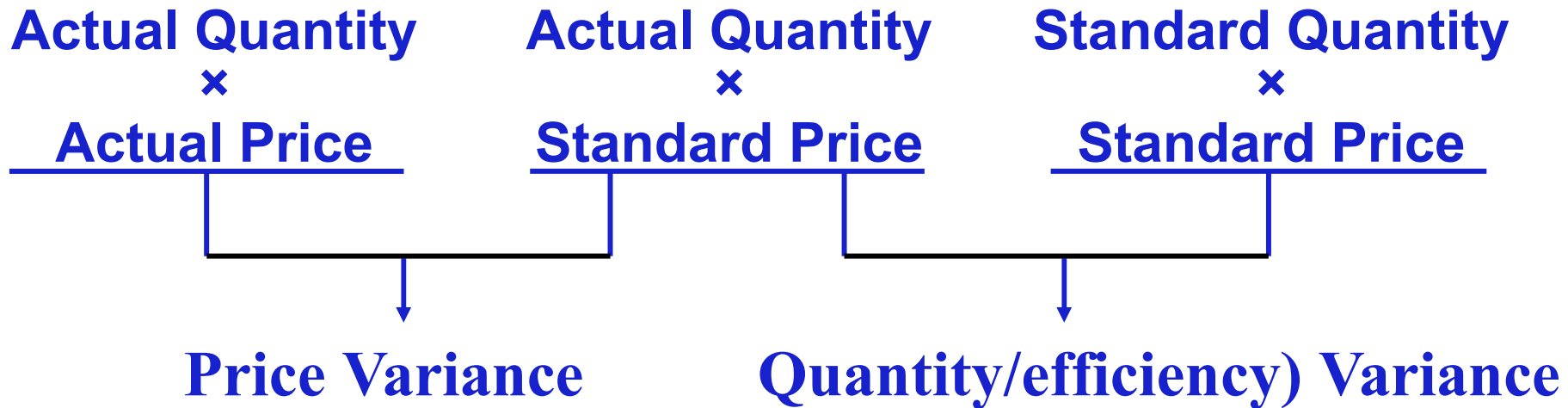
❶ The purchasing manager is responsible for raw material purchase prices and the production manager is responsible for the quantity of raw material used

❷ The buying and using activities occur at different times. Raw material purchases may be held in inventory for a period of time before being used in production.

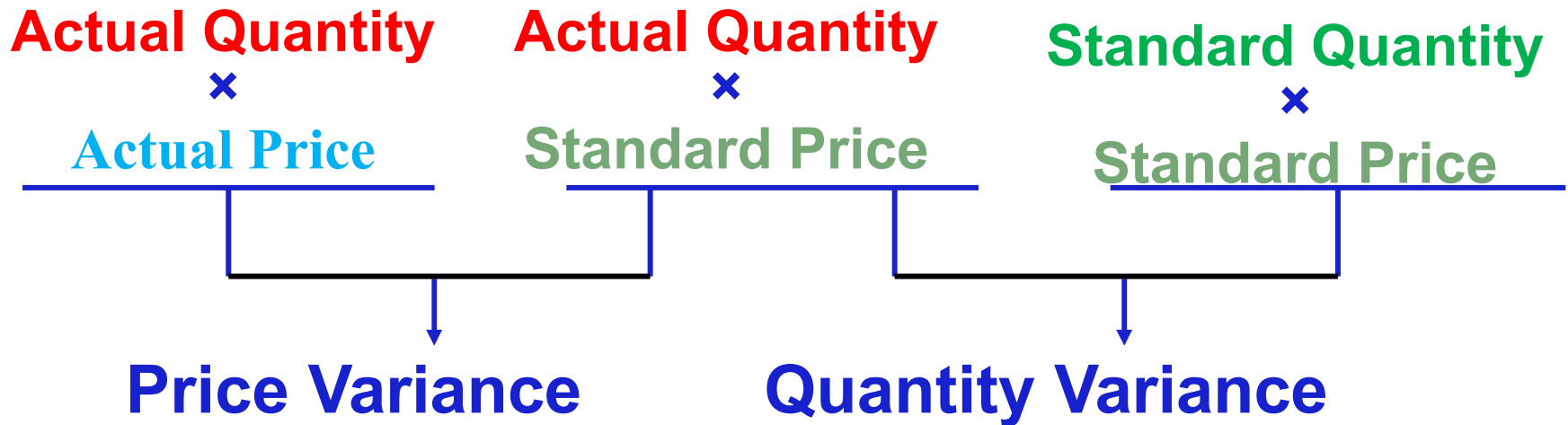
A General Model for Variance Analysis



A General Model for Variance Analysis

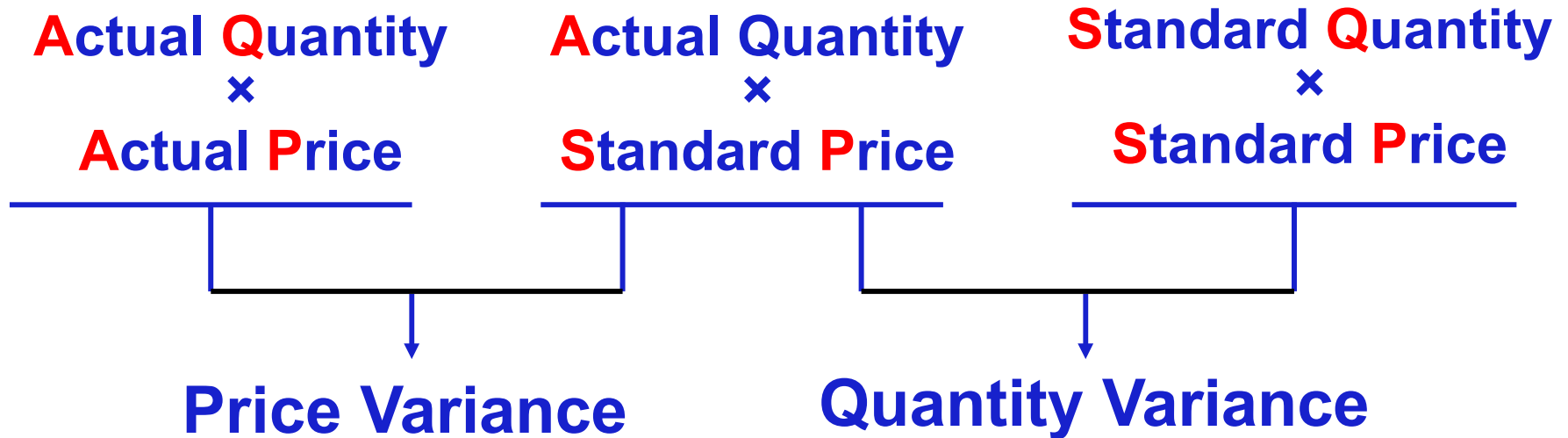


A General Model for Variance Analysis



- **Actual quantity** is the amount of direct materials, direct labor, and variable manufacturing overhead actually used.
- **Standard quantity** is the standard quantity allowed for the actual output for the period.
- **Actual price** is the amount actually paid for the for the input used
- **Standard price** is the amount that should have been paid for the input used.

A General Model for Variance Analysis



$$(AQ \times AP) - (AQ \times SP)$$

AQ = Actual Quantity

AP = Actual Price

$$(AQ \times SP) - (SQ \times SP)$$

SP = Standard Price

SQ = Standard Quantity

Direct Material Variances- Example

❑ Assume a product has the following direct material **standard** :

• **0.1 kg per a unit of product at \$5.00 per kg.**

❑ Last month, 210kg of DM were *purchased and used* to make 2,000 units of a product .

❑ Total direct material cost of the month was \$1,029.

Required:

a. Identify : SQ,AQ,SP and AP

b. Compute price variance and quantity variance?

Direct Material Variances- Example

Solution : SQ,AQ,SP& AP

❑ Standard quantity for unit produced = $0.1 * 2,000 = 200\text{kg}$

❑ Actual quantity for unit produced = 210kg

❑ Standard price of material = $\$5/\text{kg}$

❑ Actual price of material = $\$1,029 / 210\text{kg} = \4.9

Solution: Price Variance and Quantity Variance

Actual Quantity × <u>Actual Price</u> 210 kgs. × \$4.90 per kg. = \$1,029	Actual Quantity × <u>Standard Price</u> 210 kgs. × \$5.00 per kg. = \$1,050	Standard Quantity × <u>Standard Price</u> 200 kgs. × \$5.00 per kg. = \$1,000
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Price variance
\$21 favorable



Quantity variance
\$50 unfavorable

Material Variances: Using the Factored Equations

Materials price variance

$$\begin{aligned}\text{MPV} &= \text{AQ} (\text{AP} - \text{SP}) \\ &= 210 \text{ kgs} (\$4.90/\text{kg} - \$5.00/\text{kg}) \\ &= 210 \text{ kgs} (-\$0.10/\text{kg}) \\ &= \$21 \text{ F}\end{aligned}$$

Materials quantity variance

$$\begin{aligned}\text{MQV} &= \text{SP} (\text{AQ} - \text{SQ}) \\ &= \$5.00/\text{kg} (210 \text{ kgs} - (0.1 \text{ kg/parka} \times 2,000 \text{ parkas})) \\ &= \$5.00/\text{kg} (210 \text{ kgs} - 200 \text{ kgs}) \\ &= \$5.00/\text{kg} (10 \text{ kgs}) \\ &= \$50 \text{ U}\end{aligned}$$

Material Variances



How are the variances computed if the amount purchased **differs** from the amount used?



The price variance is computed **on the entire quantity purchased.**

The quantity variance is computed only **on the quantity used.**

Direct Labor Variances - Example

❑ A company has the following direct labor standard for its product

❑ **1.2 standard hours per unit at \$10.00 per hour**

❑ Last month employees actually worked 2,500 hours at a total labor cost of \$26,250 to make 2,000 units of product.

a. Identify : SQ,AQ,SP and AP

b. Compute price variance and quantity variance

Solution

- Standard quantity (hours) for unit produced = $1.2 * 2,000 = 2,400\text{hrs}$
- Actual quantity for unit produced = $2,500\text{hrs}$
- Standard price of labor = $\$10/\text{hr}$
- Actual price of labor = $\$26,250 / 2,500\text{hrs} = \10.5

Labor Variances Summary

Actual Hours
×
Actual Rate

2,500 hours
×
\$10.50 per hour

= \$26,250

Actual Hours
×
Standard Rate

2,500 hours
×
\$10.00 per hour.

= \$25,000

Standard Hours
×
Standard Rate

2,400 hours
×
\$10.00 per hour

= \$24,000

Rate variance
\$1,250 unfavorable

Efficiency variance
\$1,000 unfavorable



Labor Variances: Equations Method

Labor rate(price) variance

$$\begin{aligned}\text{LRV} &= \text{AH} (\text{AR} - \text{SR}) \\ &= 2,500 \text{ hours } (\$10.50 \text{ per hour} - \$10.00 \text{ per hour}) \\ &= 2,500 \text{ hours } (\$0.50 \text{ per hour}) \\ &= \$1,250 \text{ unfavorable}\end{aligned}$$

Labor efficiency(quantity) variance

$$\begin{aligned}\text{LEV} &= \text{SR} (\text{AH} - \text{SH}) \\ &= \$10.00 \text{ per hour } (2,500 \text{ hours} - 2,400 \text{ hours}) \\ &= \$10.00 \text{ per hour } (100 \text{ hours}) \\ &= \$1,000 \text{ unfavorable}\end{aligned}$$



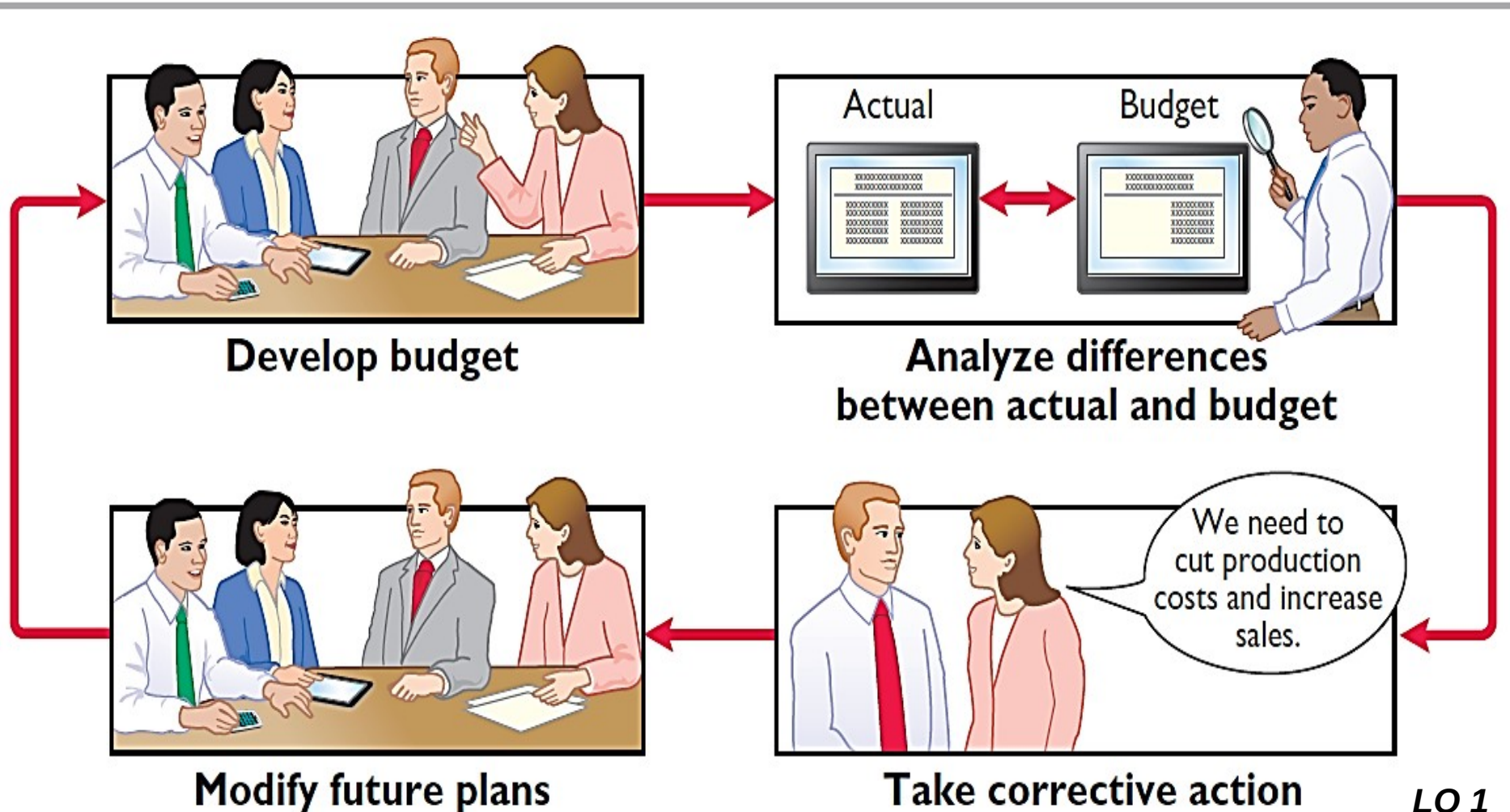
Learning Objectives- II

*Static budget and performance report using static(Master)
budget(Level-1 variance analysis)*

Budget as a tool to Control Business Activities

How Do Managers Use Budgets to Control Business Activities?

Budgetary control involves the following activities.



How Do Managers Use Budgets to Control Business Activities?

- Managers use budgets for planning and controlling business activities.
- The master budget focuses on the planning step.
- The controlling step focuses on the decisions managers make **during and after the budgeting period**, based on the actual results
- Actual results are compared with budgets and differences are investigated and analyzed.
- Corrective measures are taken based the anylysis and future budget is modified accordingly.

Learning Objective I

Static Budget and Performance Report

Static budget and Static-Budget Variance

- **Static budget** – a budget prepared for only one level of activity
 - It is based on the level of output planned at the start of the budget period.
 - The master budget is a static budget.
- **Static-Budget Variance (Level 1 variance)**
 - The difference between the actual operating income and the static budget operating income amount
 - This Variance can be Favorable or Unfavorable variance

Static-Budget Variance cont...

- Favorable (F) variance exists if an actual operating income is greater than budgeted operating income:
 - Actual revenue $>$ Budgeted revenue
 - Actual expense $<$ Budgeted expense
- Unfavorable (U) variance exists if an actual operating income is less than budgeted operating income :
 - Actual revenue $<$ Budgeted revenue
 - Actual expense $>$ Budgeted expense

Static Budgets Performance Report

- Is an income statement report prepared by comparing Budgeted income statement and actual income statement.
- The report also shows Static Budgets variance by comparing the two reports
- Static budget variance is also called **level 1 variance**
- Answers a question of “How much were we off?”

Example: Performance Reports Using Static Budgets

- Assume the following **budgeted and Actual data** for Green Manufacturing plc for the year ended December 31,2020.

Item	Actual Result	Static(Master)Budget
<i>Sales (Q)</i>	<i>10,000 units</i>	<i>8,000 units</i>
<i>Unit selling price</i>	<i>\$12.1</i>	<i>\$12</i>
<i>Unit variable cost</i>	<i>\$ 8.6</i>	<i>\$8</i>
<i>Total annual Fixed Cost</i>	<i>\$19,000</i>	<i>\$20,000</i>

- Required:** Prepare Static Budgets Performance Report

Static Budgets Performance Reports

Green Plc Static Budget Performance report For the year ended decemeber31,2020			
Particulars	Actual	Static/ Master Budget	Static/Master Budget /Variances
<i>Units</i>	<i>10,000 units</i>	<i>8,000 units</i>	<i>2,000 units F</i>
Sales	\$121,000	\$96,000	25,000 F
Variable Expenses	86,000	64,000	22,000U
Contribution Margin	35,000	32,000	3,000 F
Total Fixed Expenses	19,000	20,000	1000 F
Operating Income	<u>16,000</u>	<u>12,000</u>	<u>4,000 F</u>

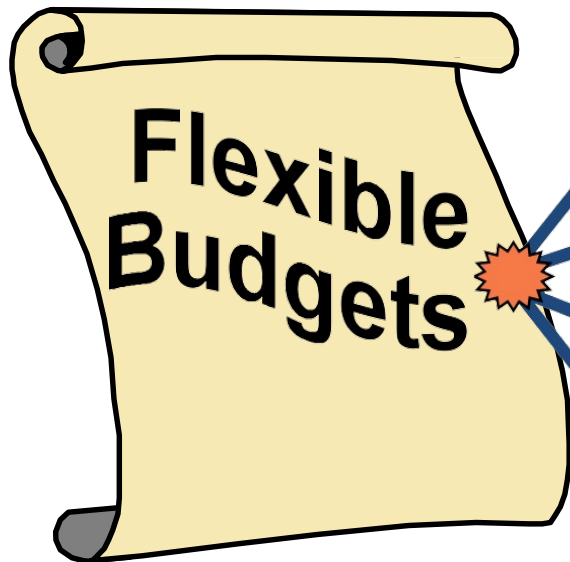
→ Level 1

Learning Objective 2



- Level 2 variance analysis
- Prepare flexible budgets
- Performance reports using static and flexible budgets

Purpose of Flexible Budgets



Show revenues and expenses that should have occurred **at the actual level of activity.**

May be prepared for any activity level in the relevant range.

Reveal variances due to rate(price) and due to volume (quantity)

Improve performance evaluation (gives more detail variance analysis)

Preparing Flexible Budgets

- **Flexible budget**

- Is prepared for several different volume levels within a relevant range(within firms capacity)
- A flexible budget is prepared at the end of the period when the actual output is known
- shifts budgeted revenues and costs up and down based **on actual operating results (activities)**
- Represents **a combination** of actual activities(units) and budgeted dollar amounts(rate)
- Will allow analysis of Levels 2 and 3 variances
- Answers the question: “**Why were we off?**”

Preparing Flexible Budgets

- A flexible budget is prepared using budgeted rate and the actual output
- Flexible budget revenue is budgeted selling price by actual quantity
- Flexible budgets separate variable costs from fixed costs;
- Flexible budgets variable cost is computed by budgeted variable rate by actual activity
- Fixed cost flexible budget is budgeted fixed cost .
- Therefore, to prepare a flexible budget, we need to know:
 - Budgeted selling price per unit and budgeted Variable cost per unit
 - Budgeted Total fixed costs
 - Different volume levels within the relevant range

Example: Preparing Flexible Budgets

- Prepare Flexible budget for Green Manufacturing plc using the following data using 5000,8000 and 10,000 level of activity .

Item	Actual Result	Static(Master) Budget
Output(Q)	10,000 units	8,000 units
Unit selling price	\$12.1	\$12
Unit variable cost	\$ 8.6	\$8
Total Fixed Cost	\$19,000	\$20,000

Preparing Flexible Budgets

**Green Manufacturing Plc
Flexible Budget
For the year ended december31,2020**

<i>Units</i>	<i>5,000 units</i>	<i>8,000 units</i>	<i>10,000 units</i>
Sales	\$60,000	\$96,000	\$120,000
Variable Expenses	40,000	64,000	80,000
Contribution Margin	20,000	32,000	40,000
Fixed Expenses	20,000	20,000	20,000
Operating Income	<u>0</u>	<u>12,000</u>	<u>20,000</u>

Flexible Budget and Variance Analysis

Level 2 analysis

- Managers want to know *why* a variance occurred.
- Total budget (static) variance is the sum of

1. Flexible budget variance

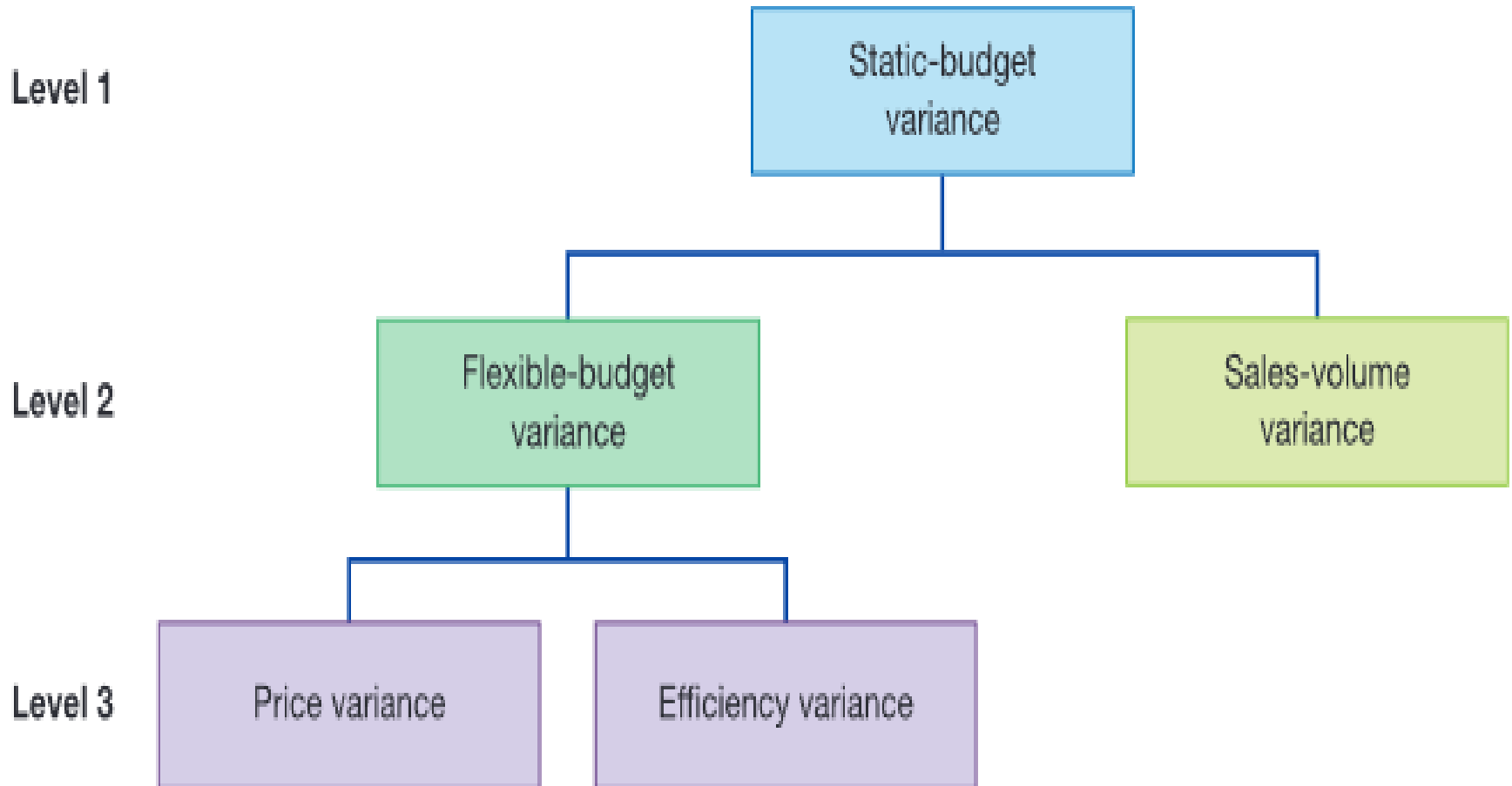
❖ is the difference between actual results and the flexible budget.

2. Sales volume variance

❖ is the difference between the flexible budget and the static budget.

- *Flexible budget is prepared using budgeted rate and actual output*

Level(detail) of Variance Analysis



Flexible Budget and Variance

Level 2 analysis

Actual Results

*Actual rate @
actual outputs*

Flexible Budget

*Budgeted rate @ actual
outputs*

Static Budget

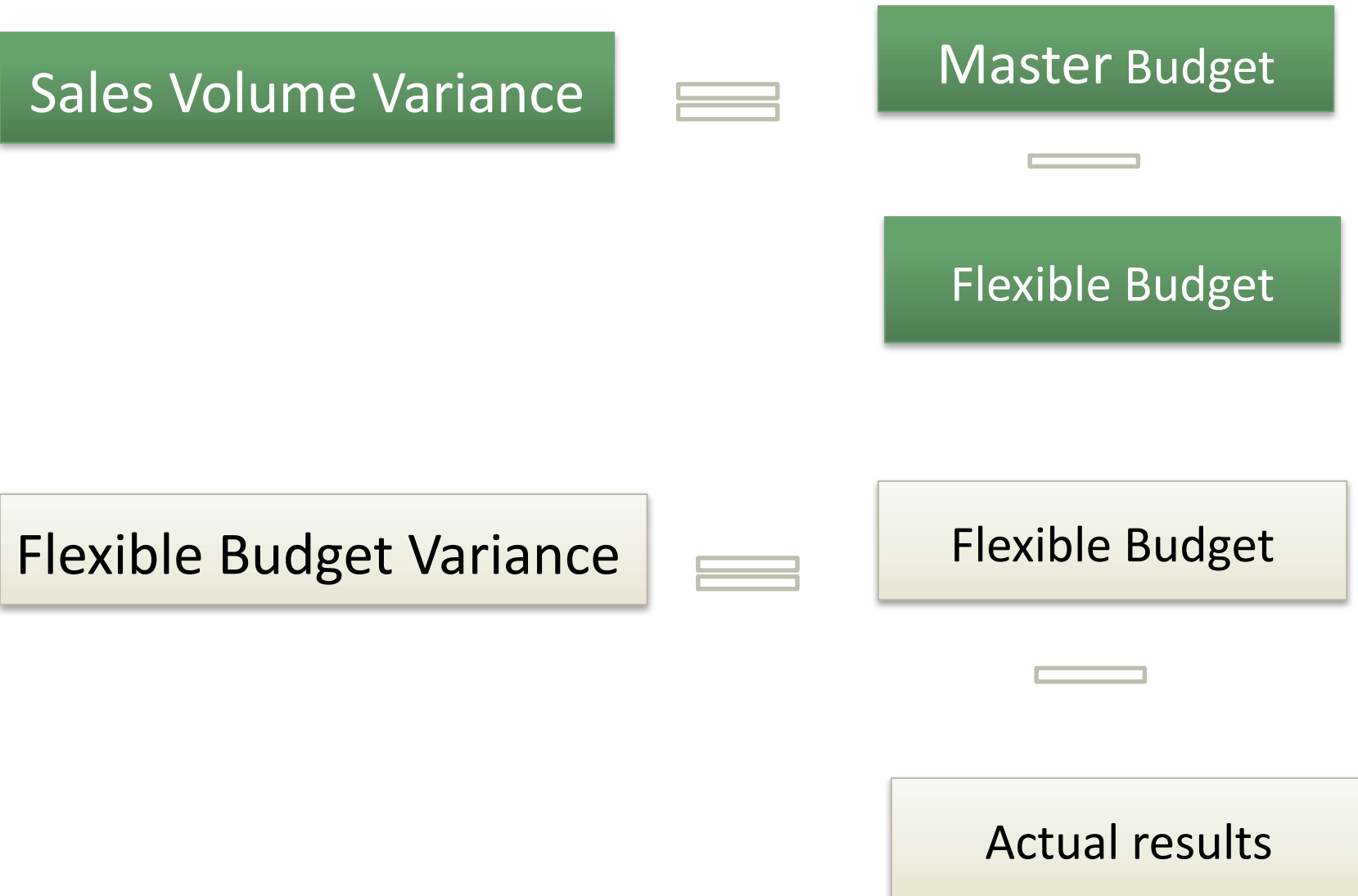
*Budgeted rate @
budgeted outputs*

Flexible Budget Variance

Sales Volume Variance

Static Budget Variance

Flexible Budget and Variance Level 2 analysis



Example-Flexible Budget and Variance Report

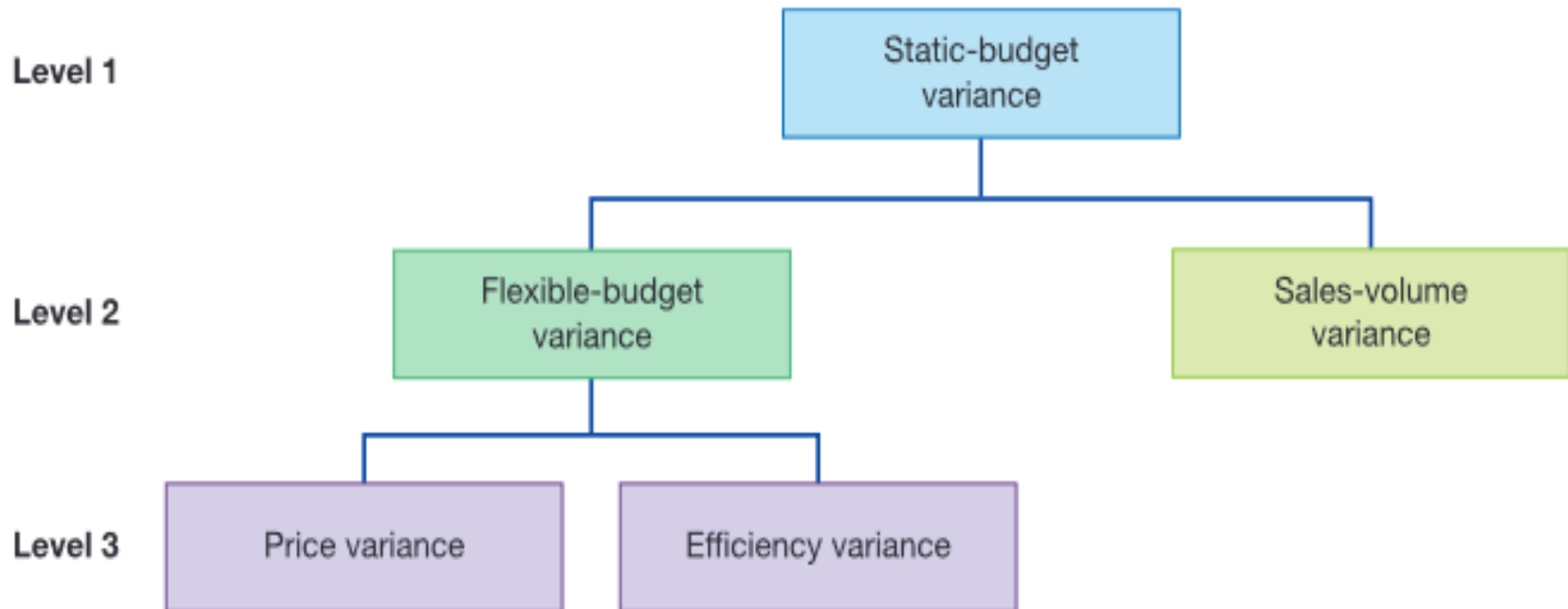
- Example prepare Flexible Budget and Variance report for Green Plc based on the data

Green Manufacturing Plc
Flexible Budget variance report
For the year ended december31,2020

	Actual Result (1)	Flexible Budget Variance = 1-2	Flexible Budget (2)	Sales Volume Variance =2-3	Master/Static budget(3)
Units	10,000	0	10,000	2000F	8,000 units
Sales	\$121,000	1000F	\$120,000	24,000F	\$96,000
Variable Expenses	86,000	6,000U	80,000	16,000U	64,000
Contribution Margin	35,000	5000U	40,000	8000F	32,000
Fixed Expenses	19,000	1000F	20,000	0	20,000
Net income	16,000	4,000U	20,000	8,000F	12,000
Flexible budgets variance \$4,000 U				Sale volume variance \$8,000 F	
Static budget variance \$4,000F					

Level 3 Variances analysis

Price Variance and efficiency Variance
for Direct Material , Direct labor and Over head costs



Level 3 Variances analysis

- Is the detail analysis of flexible budget variance
- Management needs information about the factors causing a cost variance
- Cost variance can be resulted from:
 - the input (material or labor) used to manufacture a quantity of output(usage)----**efficiency**
 - amount paid to acquire the input (material or labor)- **price**
- Therefore, we either paid more or less than planned for inputs (price variance) or
- used more or less than planned input (efficiency variance)
- Hence, cost (flexible budget) variance is the sum of price variance and efficiency variance.**

Price Variance

- Is the difference(budget vs. actual) in price of an input
- Measures how well the business keeps unit costs within standards

Efficiency (or Quantity) Variance

- Is the difference(budget vs. actual) quantity of an input used for an output.
- Measures how well the business uses its materials or human resources

Cost Variances

$$\text{Price Variance} = \left\{ \begin{array}{l} \text{Actual Price} \\ \text{Of Input} \end{array} - \begin{array}{l} \text{Budgeted Price} \\ \text{Of Input} \end{array} \right\} \times \begin{array}{l} \text{Actual Quantity} \\ \text{Of Input} \end{array}$$

$$\text{Efficiency Variance} = \left\{ \begin{array}{l} \text{Actual Quantity} \\ \text{Of Input Used} \end{array} - \begin{array}{l} \text{Budgeted Quantity of Input} \\ \text{Allowed for Actual Output} \end{array} \right\} \times \begin{array}{l} \text{Budgeted Price} \\ \text{Of Input} \end{array}$$

Illustration Example

Assume the following Data for KK Plc

	Product Cost		Qt(hrs) per output	Price per unit(hr)	Cost per Unit of finished good
Budgeted Data (Standard)	DMC		1unit	\$1.75	\$1.75
	DLC		0.25hr	\$12	\$3
	Variable OH Cost		0.25 DL hr	\$3	\$0.75
	Fixed OH Cost		0.25 DL hr	\$2	\$0.5
	<i>Production and sales</i>			<i>50,000units</i>	
	<i>DL hours</i>			<i>12,500 hrs</i>	
Actual Data	DMC	\$104,000	1.25units	\$1.6	\$2
	DLC	\$145,600	0.2hr	\$14	\$2.8
	Variable OH Cost 30,160		0.2hr	\$2.9	\$0.58
	Fixed OH Cost \$23,920		0.2hr	\$2.3	\$0.46
	<i>Production and sales</i>			<i>52,000units</i>	
	<i>DL hours</i>			<i>10,400 hrs</i>	

Required:

- a. Compute Flexible Budget Variance*
- b. Compute Price and efficiency(quantity) Variance component of flexible budget variance*

a. Flexible Budget variance

Green Manufacturing Plc Flexible Budget variance report For the year ended december31,2020			
	Actual Result (1) Q= 52,000	Flexible Budget Variance = 1-2	Flexible Budget (2) Q=52,000
Direct Material Cost	\$104,000	13,000U	\$91,000(a)
Direct Labor Cost	\$145,600	10,400F	\$156,000(b)
Variable Overhead Cost	\$30,160	8,840F	\$39,000(c)
Fixed Overhead Cost	\$23,920	2,080F	\$26,000(d)
Total Product cost Flexible Budget Variance		8,320F	
a=52,000 1.75, b=52,000*3, c= 52,000*0.75,d=52,000*0.5			

Price and Efficiency Variance

- Direct Material Cost, FBV=\$13,000U**

$$\text{Price Variance} = \left\{ \begin{array}{c} \text{Actual Price} \\ \text{Of Input} \end{array} - \begin{array}{c} \text{Budgeted Price} \\ \text{Of Input} \end{array} \right\} \times \begin{array}{c} \text{Actual Quantity} \\ \text{Of Input} \end{array}$$

$$\text{Price Variance} = (1.6 - 1.75) \times 65,000 \text{ units} = \$9,750\text{F}$$

$$\text{Efficiency Variance} = \left\{ \begin{array}{c} \text{Actual Quantity} \\ \text{Of Input Used} \end{array} - \begin{array}{c} \text{Budgeted Quantity of Input} \\ \text{Allowed for Actual Output} \end{array} \right\} \times \begin{array}{c} \text{Budgeted Price} \\ \text{Of Input} \end{array}$$

$$\text{Efficiency Variance} = (65,000 \text{ units} - 52,000 \text{ units}) \times \$1.75 = \$22,750\text{U}$$

Price and Efficiency Variance

- **Direct Labor Cost, FBV=\$10,400F**

$$\text{Price Variance} = \left\{ \begin{array}{c} \text{Actual Price} \\ \text{Of Input} \end{array} - \begin{array}{c} \text{Budgeted Price} \\ \text{Of Input} \end{array} \right\} \times \begin{array}{c} \text{Actual Quantity} \\ \text{Of Input} \end{array}$$

$$\text{Price Variance} = (14 - 12) \times 10,400 \text{ hrs} = \$20,800\text{U}$$

$$\text{Efficiency Variance} = \left\{ \begin{array}{c} \text{Actual Quantity} \\ \text{Of Input Used} \end{array} - \begin{array}{c} \text{Budgeted Quantity of Input} \\ \text{Allowed for Actual Output} \end{array} \right\} \times \begin{array}{c} \text{Budgeted Price} \\ \text{Of Input} \end{array}$$

$$\text{Efficiency Variance} = (10,400 \text{ units} - 13,000 \text{ hrs}) \times \$12 = \$31,200\text{F}$$

Price and Efficiency Variance

- Variable OH Cost, FBV=\$8,840F

$$\text{Price Variance} = \left\{ \begin{array}{c} \text{Actual Price} \\ \text{Of Input} \end{array} - \begin{array}{c} \text{Budgeted Price} \\ \text{Of Input} \end{array} \right\} \times \begin{array}{c} \text{Actual Quantity} \\ \text{Of Input} \end{array}$$

$$\text{Price Variance} = (\$2.9 - \$3) \times 10,400 \text{ hrs} = \$1,040\text{F}$$

$$\text{Efficiency Variance} = \left\{ \begin{array}{c} \text{Actual Quantity} \\ \text{Of Input Used} \end{array} - \begin{array}{c} \text{Budgeted Quantity of Input} \\ \text{Allowed for Actual Output} \end{array} \right\} \times \begin{array}{c} \text{Budgeted Price} \\ \text{Of Input} \end{array}$$

$$\text{Efficiency Variance} = (10,400 \text{ units} - 13,000 \text{ hrs}) \times \$3 = \$7,800\text{F}$$

Price and Efficiency Variance

- Fixed OH Cost, FBV=\$2,080F

$$\text{Price Variance} = \left\{ \begin{array}{c} \text{Actual Price} \\ \text{Of Input} \end{array} - \begin{array}{c} \text{Budgeted Price} \\ \text{Of Input} \end{array} \right\} \times \begin{array}{c} \text{Actual Quantity} \\ \text{Of Input} \end{array}$$

$$\text{Price Variance} = (\$2.3 - \$2) \times 10,400 \text{ hrs} = \$3,120\text{U}$$

$$\text{Efficiency Variance} = \left\{ \begin{array}{c} \text{Actual Quantity} \\ \text{Of Input Used} \end{array} - \begin{array}{c} \text{Budgeted Quantity of Input} \\ \text{Allowed for Actual Output} \end{array} \right\} \times \begin{array}{c} \text{Budgeted Price} \\ \text{Of Input} \end{array}$$

$$\text{Efficiency Variance} = (10,400 \text{ units} - 13,000 \text{ hrs}) \times \$2 = \$5,200\text{F}$$

Green Manufacturing Plc
Price and Efficiency variance report
For the year ended december31,2020

	Price Variance	Efficiency Variance	Flexible Budget Variance
Direct Material Cost	\$9,750F	\$22,750U	13,000U
Direct Labor Cost	\$20,800U	\$31,200F	10,400F
Variable Overhead Cost	\$1,040F	\$7,800F	8,840F
Fixed Overhead Cost	\$3,120U	\$5,200F	2,080F
<i>Total</i>	<i>13,130U</i>	<i>21,450F</i>	<i>8,320F</i>

Exercise

	Flexible Budget	Variances Price	Efficiency
Material A	\$20,000	\$1,000 U	\$1,200F
Material B	30,000	500 F	800U
Material C	40,000	1,400 U	1,000F

Requerid:

Compute actual DMC of Material A,B and C

Who is responsible for the variances?

Responsibility for Material Variances

Materials Quantity Variance



Production Manager

Materials Price Variance



Purchasing Manager

The standard price is used to compute the quantity variance so that the production manager is not held responsible for the purchasing manager's performance.

Responsibility for Material Variances

I am not responsible for this unfavorable material quantity variance.

You purchased cheap material, so my people had to use more of it.



Your poor scheduling sometimes requires me to rush order material at a higher price, causing unfavorable price variances.



Possible Causes of Material Cost Variances

Usage Related Issues (efficiency)

- Purchase and use of sub-standard material
- Personnel manager hired under skilled workers
- Maintenance department did not properly maintain machines.

Price Related Issues

- Fail to purchase anticipated materials at appropriate price
- Poor purchase scheduling may rush order material at a higher price

Possible Causes of Labor Cost Variances

- Production managers are usually held accountable for labor variances

Time Related Issues(efficiency)

- Change in design and quality standard
- Low Motivation
- Poor working conditions
- Improper scheduling/placement of labor
- Inadequate Training
- Quality of production supervision.

Rate(price) Related Issues

- Increments / high labor wages
- Overtime
- Labor shortage leading to higher rates
- Union agreement

Possible Causes of overhead Cost Variances

Fixed Overhead Cost

- Under or over estimation of fixed overheads
- Fall in demand/ under capacity
- Breakdowns /Power Failure

Variable Overhead Cost

- Inflation
- Lack of planning
- Lack of cost control

Possible Causes of Sales volume Variance

- Change in Market size
- Change in Market share

Summery- Who is responsible for the variances?

Cost	Price Variance	Efficiency variance
Direct Materials	Responsibility: Purchasing managers	Responsibility: Production managers & engineers
Direct Labor	Responsibility: Human resource managers	Responsibility: Production managers & engineers
Manufacturing Overhead	Responsibility: Production managers	

Variance Analysis and Management by Exception

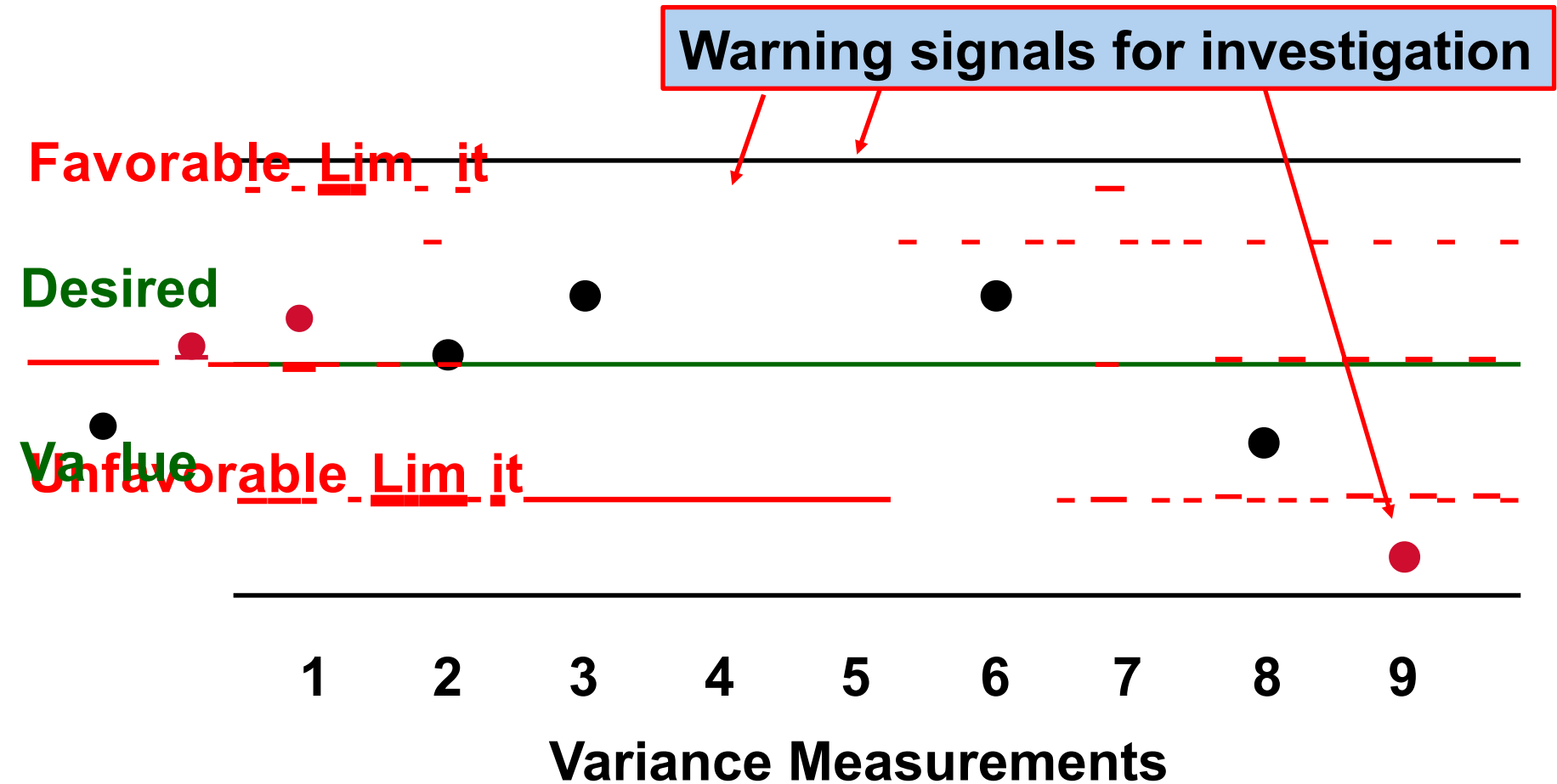


How do I know which variances to investigate?



Larger variances, in dollar amount or as a percentage of the standard, are investigated first.

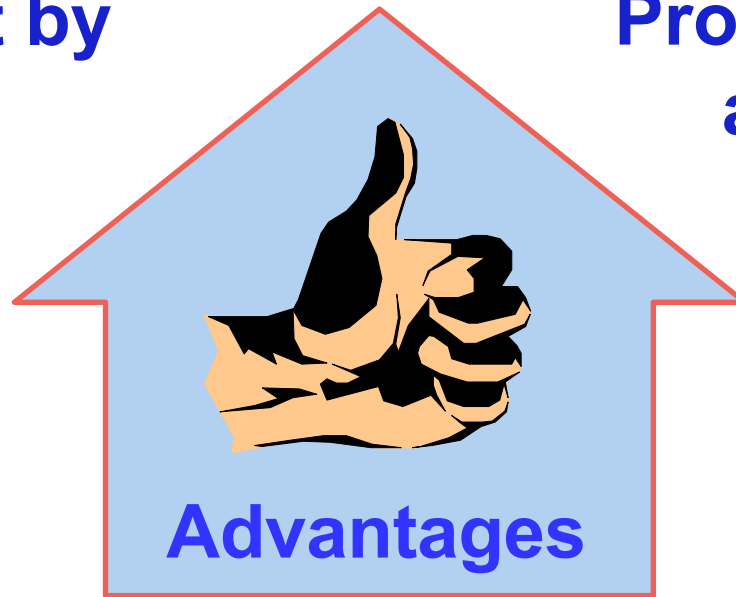
A Statistical Control Chart



Advantages of Standard Costs

**Management by
exception**

**Promotes economy
and efficiency**



**Simplified
bookkeeping**

**Enhances
responsibility
accounting**

Measures taken to be on the variances

- Management should correct problems that caused unfavorable variances and
- Adopt and reward the practices that resulted in favorable variances

The End